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COUNTERING PSEUDOLINGUISTIC CLAIMS

The Ethical Challenges Facing Forensic Linguists

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Introduction

Ethics is concerned with morality and norms for behavior, or, in basic terms, what we ought to do or not do. Given that forensic linguistics deals both with the letter of the law and with the prosecution of transgressions of the law, ethical issues form a steady undercurrent to everything forensic linguists do as professionals and academics. However, there are also ethical challenges they face that are related to the actions of others outside of the profession—persons who (ab)use linguistics to make unsubstantiated claims about the intelligence that can be extracted from language data. This chapter reviews some of the pseudolinguistic schemes that are marketed to law enforcement, judicial systems, and the general public to convince them that simple, easy-to-use tools can reveal who is lying and who is truthful, who is guilty and who is innocent. We deconstruct three typical claims from operatives in this marketplace regarding language and linguistics to demonstrate how they are "epistemically irresponsible" (Heffer 2020: 178). This chapter argues that the marketing of such pseudolinguistic methods in the realm of legal prosecutions poses a significant ethical challenge to forensic linguists who not only seek to debunk false claims but also offer informative and understandable, evidence-based counter-narratives that are rooted solidly in linguistic science.

This chapter cannot, and does not attempt to, represent the full breadth and extent of pseudolinguistic offers on the market. It would be impossible to account for all those being promoted outside of the public sphere. The same is true of the many previously debunked methods that are being peddled under new names (or acronyms, as the fashion would have it). One of the pragmatic challenges that faces the forensic linguist is to remain vigilant for new developments that may cause harm. The difficulty in keeping abreast of the latest pseudoscientific developments may be greater in countries with few forensic linguists to share the burden of discovering and countering misleading claims about language in the legal realm. It is therefore important for forensic linguists to work as an international research community and share information about pseudolinguistic methods. By the same token, forensic linguists must warn practitioners and the public alike about the potential risks to the dispensation of justice if such legitimate linguistic methods are improperly used or illegitimate methods are used at all.

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Pseudolinguistic Methods and Language Evidence

Language is used for legal (and illegal) purposes in myriad ways, and forensic linguistics is continually branching out to cover an increasing number of them. For some time now, linguistic expertise has been applied to improve investigative and judicial procedures and to offer expert testimony in civil and criminal cases. However, there are other areas where language and the law clearly intersect but the science of linguistics has (as yet) had little to no influence on current practices. This state of affairs leaves such areas open to linguistically untrained actors who may or may not genuinely believe that they understand how language works. For instance, in the area of policing, interrogations are still being conducted with coercive techniques for verbal "deception detection" and "credibility analysis". What many of these law enforcement agencies may not realize is that none of these methods is based on a real understanding of linguistic theory. Although many champions of these methods frequently use linguistic terminology, theory, or results, the fact is, they often do so in a faulty and misleading manner. As will be shown in the following section, this misuse of linguistics can have significant negative consequences.

Police Interviews

Internationally, there are many competing approaches to police interviews on offer today. To distinguish them, a dividing line is often drawn between two main groups: (1) "accusatorial" (e.g., Meissner et al. 2012) or "confession-based" (e.g., Davis and Leo 2013) methods; and (2) "information-gathering" or "cognitive" methods (Memon and Gawrylowicz 2019). To the first group belongs the often-discussed "Reid Technique" (Inbau and Reid 2011 and later; Jane and Buckley 1999) and to the other belong the English-Welsh *PEACE* model (where the abbreviation stands for *Planning and Preparation, Engage and Explain, Account Clarification and Challenge, Closure, Evaluation*) (Clarke and Milne 2001; Cherryman and Bull 2000) and the Norwegian *KREATIV* model (*Kommunikasjon "Communication", Rule of law, Ethics and empathy, Active consciousness, Trust through openness, and Information, Verified through science*) (Fahsing and Jakobsen 2017). Studies in the United Kingdom and elsewhere have demonstrated that intelligence gathered from cognitive interviewing is richer and more reliable than intelligence gathered from confession-based methods (e.g., Meissner et al. 2012; Memon, Meissner, and Joanne Fraser 2010).

The Reid Technique, named after its primary inventor John E. Reid, is widely used throughout the globe. According to the guidelines outlined in the "Nine Steps of Interrogation", a prerequisite for using this method is that "the investigator is reasonably certain of the suspect's guilt" (Inbau and Reid 2011: 5). Such a presumption of guilt need not rely on independent investigatory evidence but can be generated by the initial non-accusatory "Behavior Analysis Interview". As has been shown repeatedly, the indicators of truth and falsity suggested for that pre-interrogation interview are scientifically unsubstantiated and operate upon the reductionist notion that only "guilty suspects" lie (Vrij et al. 2022; Heydon 2019; Hirsch 2014). Unsurprisingly, there are plenty of examples of false confessions that have resulted from the deployment of the Reid Technique against innocent people (see, e.g., Hirsch 2014; Gudjonsson and Pearse 2011; Moore and Fitsimmons 2011; Ofshe and Leo 1997).

The corporation behind the Reid Technique, Reid & Associates, Inc., is very aware of the decades of criticisms that have been lodged against their methods, as evidenced by the long list of responses on their "Critics Corner". Nevertheless, in an examination of the adjustments made to the Reid Technique over the years, Hirsch (2014) finds that its tactics still carry grave risks of "breaking" innocent people and making them confess to crimes they did not commit. In the fol-

lowing section, examples are given to demonstrate some of the primary problems with the Reid Technique and how these issues are directly related to fundamental misunderstandings and misrepresentations about language and language users.

Abuse of Language within the Reid Technique

A simple look at the training programs offered by Reid & Associates (https://reid.com n.d.) gives cause for concern. Within "The Reid Nine Steps of Interrogation" participants will learn what is summarized as "Handling Denials", "Overcoming Objections", "Challenging the Suspect's Values and Traits", and "Committing the Suspect to the Crime". Given that these interrogation tactics are dispersed throughout the nine steps, they leave precious little room for suspects to change the interrogators' decision that they are guilty.

The concern that the Reid Technique can be used for unethical coercion is heightened when the actual content of the courses is examined. The excerpt below is taken directly from the Reid Technique's "Step Three: Handling Denials":

> A guilty suspect will oftentimes introduce his denials by asking permission to speak with such phrases as, "May I say something?" or "If you just let me say one thing…" Were the investigator to allow the suspect to make his full statement, each of these introductory remarks would be followed by the denial, "I didn't do it." Therefore, whenever a suspect asks to speak during this phase of the interrogation, the investigator should respond by saying, "Andy, just hold on for a minute. Let me explain to you exactly what we know."
>
> *(Reid & Associates n.d.)*

In this excerpt, it is not only taken for granted that a polite request to speak will inevitably be followed by a denial of guilt but also assumed that such a combination of speech acts will only occur with guilty suspects. There is no consideration of the variation exhibited in politeness formulae (e.g., Schneider and Placencia 2017) or their interaction with the asymmetrical and stressful situation of being subjected to a high-stakes police interrogation (e.g., Archer 2017), nor of the possible influence of cross-cultural misunderstandings (e.g., Sifianou and Blitvich 2017). It is simply assumed that, aside from an interviewee's guilt, there are no other reasons for someone in police custody to preface their questions with variants of "May I say something?" Later in the explanation of Step Three, this excerpt contrasts the guilty suspect's behavior with that of the innocent:

> The innocent suspect, however, will generally not ask permission to speak when he wants to make a statement. Instead, he will, without any display of etiquette, promptly and unequivocally state that he had absolutely nothing to do with the theft. Innocent denials will strengthen in time, and the suspect will begin to assert control over the interrogation. In the event of such a development, the investigator must begin to modify the intensity of his position and begin a probing search for some other information such as the suspect's suspicions about who may have committed the crime. This will help explain to the suspect why he was questioned. It is very rare for an innocent suspect to move past this denial state. He remains steadfast in the assertion of his innocence. The sincerity with which he presents his case is a clear indication of his truthfulness.
>
> *(Reid & Associates n.d.)*

Here, it is unequivocally asserted that innocent people will maintain their innocence over time, despite repeated accusations by law enforcement. This is consistently countered by experiments in social psychology (e.g., Kassim 2015; Kassim and Kiechel 1996).

Criticisms of the Reid Technique are not limited to their questionable interpretations of interviewee language and behavior. It has been frequently reported that law enforcement employing this method have subjected suspects to hours-long interrogations in which officers lie to suspects about available evidence against them. Reid & Associates deny that such tactics are promoted in their programs (Buckley 2022) but there are multiple reports that counter those claims. An excellent example comes from a case that took place in New York in 2014 (https://www.ei-ie.org/en/item/19574:denmark-and-usa-malthe-thomsen-dismissed-of-all-charges). A young Danish man, Malthe Thomsen, was accused by a co-worker of sexual misconduct against children at the pre-school where he was interning. Mr. Thomsen was put through a seven-hour interrogation during which the officers made repeated reference to video evidence allegedly proving his guilt. After six hours he gave a written confession, which he later retracted, claiming coercion. Charges were later dropped as the video showed no evidence of sexual misconduct and the co-worker was determined to have acted in revenge after she was fired from the pre-school.

In short, confession-based methods of police interrogation carry a high risk of false confessions based on misrepresentations of what suspect language looks like.

Deception Detection

So-called "deception detection" is a highly appealing concept within law enforcement since evaluating the validity of witness statements is crucial for successful investigations. Disproving the faulty claims of witnesses risks wasting valuable resources that could have been invested in more promising avenues of investigation. At the same time, failing to uncover untruth carries a risk of convicting innocent people and letting the real perpetrators go free. To minimize these risks, any tool that offers a simple and supposedly foolproof method of detecting untrue statements will understandably be very attractive to investigators.

SCAN

One of the oft-discussed methods of deception detection is the misleadingly named *Scientific Content Analysis*, or *SCAN*. It was developed in the 1980s by Avinoam Sapir, a polygraph examiner. Sapir designed the tool based on the premise "that the structure and contents of a subject's statement reveal whether there's an attempt at deception" (LSI Laboratory for Scientific Interrogation, Inc.). Like many other approaches to veracity assessment, SCAN is partially based on the "Undeutsch Hypothesis". Named for its inventor, Udo Undeutsch, this hypothesis proposes that false statements differ in content and quality from truthful statements because they are not based on actual memories (Vrij 2015). Based on this premise, SCAN works by having a subject write down their account of events in a statement, which the SCAN-trained officer then analyzes. This analysis is done by matching the statement against a set of linguistic criteria that indicate either truthfulness or deception.

However, one of the main criticisms of SCAN that is repeatedly reported in the scientific literature is its lack of a defined set of linguistic criteria against which a statement should be judged. For example, Bogaard et al. (2013b: 2) report that when one of the authors attended the SCAN course, "no checklist of criteria is offered, nor does the SCAN manual contain such a list". In other words,

the tool is not standardized, meaning that the results of a SCAN analysis will vary according to the person conducting the analysis. To find out what criteria are most frequently used by SCAN-trained officers, Bogaard et al. (2013b) analyzed 82 sexual abuse case statements from the Netherlands where SCAN had been applied by a police officer. They found that there are largely 12 criteria that drive SCAN analyses, 7 of which indicate truthfulness (denial of allegations, social introductions, structure of the statement, emotions, objective and subjective time, first person singular pronouns, and past tense verbs), while the remaining 5 indicate deception (change in language, spontaneous corrections, lack of conviction or memory, out of sequence, and extraneous and missing information) (Bogaard et al. 2016: 4, 2013b). Furthermore, they found that although the police officers examining these statements had received SCAN training, their inter-rater reliability was extremely low, suggesting "that the SCAN method is insufficiently developed as a forensic tool" (Bogaard et al. 2013b: 15). In fact, the available literature predominantly argues against its use given its lack of empirical support and its susceptibility to contextual bias, i.e., that surrounding circumstances may influence the analysis and interpretation of the statement (Goormans, Mergaerts, and Vandeviver 2022; Heydon 2019; Bogaard et al. 2016; Vrij et al. 2015; Bogaard et al. 2013a; Bogaard et al 2013b; for a comprehensive review see Vrij et al. 2022).

911-Call Analysis

Another questionable method of deception detection is "911-call analysis". Surfacing in the early 2000s, this method was developed by former US police officer Tracy Harpster. According to Harpster, 911-call analysis provides a reliable way for investigators to determine whether the caller is telling the truth (Harpster and Adams 2017; Harpster, Adams, and Jarvis 2009; Harpster 2006). This method offers 8 variables, all based on the Undeutsch hypothesis, that are associated with innocence, e.g., "plea for help" and "self-corrections", for example, explicitly saying "I need help" during the call, and 12 variables that are associated with guilt, e.g., offering "extraneous information" and exhibiting "inappropriate politeness", such as saying *thank you* to the 911-call operator (Harpster, Adams, and Jarvis 2009). Based on these variables, Harpster devised a scale called "Considering Offender Probability in Statements" or COPS. The purpose of this scale is to assist police in evaluating 911 calls. To date, Harpster has declined to disclose the data on which COPS is based, referencing a concern that researchers would "steal" his research to "profit" from it (Murphy 2022). As with both SCAN and the Reid Technique, research examining 911-call analysis is scarce. The few reports that have been published appear to stem from authors affiliated with the tool in question and may therefore be biased in favor of it (e.g., Elms and Keatley 2023; O'Donnell et al. 2023; Cromer et al. 2018).

All in all, existing research into deception detection methods such as SCAN and 911-call analysis shows that they are difficult to deploy consistently and that they produce conflicting and misleading results. Even more troubling, however, is the fact that there is no scientific backing for the claim that any of these methods can reliably detect whether someone is being deceptive based on linguistic features. Instead, it seems that the reported effectiveness of such tools is more a function of confirmation bias than any underlying correlation between linguistic features and deception. The continuing popularity of these tools in law enforcement would seem to be driven by a widespread lack of basic knowledge about how linguistic interactions work (Heydon 2019; Picornell 2013). By sharing their expertise and experience with law enforcement, forensic linguists can help decrease not only the continued use of dubious language analysis methods but also the potential for avoidable miscarriages of justice.

An Analysis of Pseudolinguistic Claims

Before we begin our data analysis, we must address an important issue regarding the sources for our data. While we believe it is important to identify and critique the use of pseudolinguistic methods, we also recognize the potential harm that could be caused by singling out individuals advocating such methods. Moreover, the market for deception detection and credibility analysis is constantly changing, making it quite difficult as researchers to determine exactly who is the primary proponent of a given claim. For these reasons, we have chosen not to reveal the specific sources for our data. The main focus of this study is to provide a description of the tendencies within a field that is marred by pseudoscience. Consequently, we limit ourselves to paraphrasing some of the recurring claims we have encountered throughout a year-long survey of several private companies that offer credibility assessments or deception detection services, based on tools such as the Reid Technique or SCAN, and promote their services to law enforcement, prosecution, and other authorities. Our focus here is on deconstructing the claims to show why they are problematic from a theoretical and practical viewpoint. By avoiding the naming of specific individuals or companies, we hope to encourage a broader dialogue around the need for rigorous and ethical standards in forensic linguistic practices. We believe that this approach will ultimately serve the interests of the broader forensic linguistic community. The five claims are presented below.

Claim One: The Invisibility of Function Words

Function words such as prepositions, pronouns, and adverbs are "invisible" to the brain and have little meaning because they are the "glue" that holds meaningful words together. Function words are uttered and perceived unconsciously because they are located in a different place in the brain. They are interesting when evaluating credibility because they reveal useful information about our relationships and feelings toward ourselves and others.

Note: This supposition appears to draw on Pennebaker (2013).

Claim Two: The Fixed Meaning of First Person Pronouns

The pronoun *we* can never designate a negative relationship, because *we* is a word that indicates cooperation and community. The personal pronoun *I* is a word which grants responsibility as well as revealing that the person's focus is on themselves. So, if someone says: "My wife and I went shopping" it indicates emotional attachment. But if they say: "I went shopping with my wife" it indicates distance.

Note: This supposition appears to stem from SCAN training (Heydon 2019).

Claim Three: The Suspicious Variation of Terminology

Referring to individuals inconsistently throughout a written statement, e.g., referring to your alleged rapist interchangeably as *someone*, *the guy*, *the boy*, and *the man*, is problematic and can indicate deception.

Note: This supposition also appears to stem from SCAN training (Goormans, Mergaerts, and Vandeviver 2022).

Claim Four: The Suspicious Use of Redundant Words

If someone is confident that what they say is true, they will have no need to convince their recipient of their confidence. They would just deliver the information as is, with no "superfluous" words,

that is, words that can be deleted from a sentence without making it grammatically incorrect. Superfluous words can indicate deception in, e.g., emergency calls.

Note: This supposition appears to draw partly upon studies of linguistic modality that make the point that "there is no epistemically stronger statement than a categorical [non-modalized] assertion" (Lyons 1977: 809).

Claim Five: The Unambiguous Protestation of Innocence

People mostly speak directly and unambiguously about the topic at hand, and they will say explicitly if they are innocent. In other words, if someone is falsely accused of a crime, it should not be difficult for them to deny the accusation in a clear and distinct way.

Note: This supposition appears to stem from the Reid Technique (see above).

Epistemically Irresponsible Discourse

Although the above five claims seem to be empirically driven, in the communication of them, the proponents of the claims employ (at least) two subtypes of what Heffer (2020: 182) terms epistemically irresponsible discourse pathologies, namely those of *distortion* (i.e., linguistically misrepresenting the evidence), *withholding* (i.e., omitting, evading, and even blocking counter-evidence), and *dogma* (i.e., closed-mindedness that disregards counter-evidence). Let us illustrate this by deconstructing each claim.

Claim One. Aside from the obvious contradiction that function words "have little meaning" while still somehow revealing important information about our relationships and feelings, we can identify several other problematic statements. The first sentence is partially taken (without reference) from Chung and Pennebaker (2007: 345) who write:

> We can begin to detect linguistic style by paying attention to "junk words"—those words that do not convey much in the way of content. These junk words, usually referred to as function words or particles, serve as the cement that hold content words together.

It is important to note that Chung and Pennebaker (2007) make the above statement in the specific context of analyzing linguistic style. However, the proponents of Claim One misuse this assertion and apply it to evaluating credibility.

The claim that function words are "invisible" to the brain belongs to what Heffer (2020: 182) impishly describes as a type of claim that is "evidentially worthless because its meaning is unclarifiable". The final statements that function words are "uttered and perceived unconsciously" and that they are "located in a different place in the brain" are cases of distortion by misstatement. We know from research in psycholinguistics that cognitive capacities such as speech processing are "effortless, reliable, fast, unconscious and require no instruction" (Kemmerer 2015: 111), so it is arguably irrelevant to claim that the brain perceives function words unconsciously, because *all* speech processing happens at an unconscious level. And while research has shown that different areas in the brain are indeed activated in the perception of different word classes (Diaz and McCarthy 2009), it is misleading to present this as an indication that words are "located" in separate "places in the brain" and that function words are thereby collectively good indicators of deception. Overall, Claim One is an example of how psycholinguistic research can be misrepresented for the purpose of making unsubstantiated promises about deception detection.

As has been noted previously by other research (e.g., Heydon 2019), research into linguistic markers of deception tends to be conducted by psychologists rather than linguists and is usually part of "a broader study of deceptive behaviour" in which "the explanations for the use of linguistic cues indicating deception or truthfulness are explained in relation to theories of cognition or behaviour" (Heydon 2019: 62). However, oftentimes individuals marketing specific deception detection tools will ascribe psychological motives or unconscious reasons to the linguistic choices individual speakers make, without contextualizing them within a broader cognitive or psychological theory. Let us examine this in further detail by deconstructing Claim Two.

Claim Two. In our survey, we find this claim phrased in ways that indicate the original source to be *The Secret Life of Pronouns: What Our Words Say About Us* (2013), a popular science book written by social psychologist James W. Pennebaker to demonstrate how the text-analysis tool LIWC (Linguistic Inquiry and Word Count) can be used. LIWC is a computational tool designed to examine large quantities of texts and works on the assumption that our language reflects our internal psychological states, emotions, ways of thinking, and social concerns (Pennebaker 2013). In Pennebaker's version, Claim Two was made in the context of determining the emotional state of an author over time, by computationally analyzing their written statements. But we find that it is frequently applied to analyses of spoken language in the context of deception detection, thus becoming distorted partly due to a lack of distinction between written and spoken language.

Claim Three. This claim appears to be a variation of a claim cited from SCAN training material, where an object is first referenced using a full noun phrase and then taken up again in the form of the pronoun *it* (Heydon 2019: 63). The underlying notion is that the writer inadvertently reveals an inconsistency in their narrative by shifting either between different noun phrases or a noun phrase and a pronoun. However, as Heydon (2019: 63) points out, these analyses often suffer from a lack of basic understanding of language, i.e., the use of anaphora is not a symptom of deception but rather of perfectly normal language use.

Claims Four and Five. These claims are problematic from a purely theoretical viewpoint, in part because they assume a link between psychological states and linguistic behavior using contrived examples. The assumption behind Claims Four and Five is that there is a clear distinction between "innocent" and "guilty" linguistic behavior, an assumption we recognize from both 911-call analysis and the Reid Technique, and for which there is no empirical evidence.

For example, the statements that "most people speak directly and unambiguously about the topic at hand" and that truth-tellers "would deliver the information as it is, with no extra add-ons" conflict with decades of linguistic research in fields such as conversation analysis (CA) about how the practices of spoken language "fit together in highly intricate ways" (Sidnell 2010: 1). Speakers design their turns specifically for their interlocutor, and even more specifically they design it to align to the project undertaken in the previous turn (Schegloff 2007). In other words, "the response (…) that promotes the accomplishment of the activity underway is, typically, 'preferred'" (Sidnell 2010: 77). *Preference* here does not refer to the psychological preferences of the speakers in a conversation, but to the structural preferences that organize much of talk-in-interaction. Dispreferred responses are often delayed by inter-turn gaps, turn initial delays, insertions of whole sequences like accounts, palliatives, tokens of appreciation, apologies, etc., that are oriented toward mitigating the valence of a dispreferred turn. These structural preferences are normative, meaning that they are dictated by the social norms we cultivate and care about. However, in the context of credibility assessment they are often interpreted as being indicative of deception, perpetuating the notion that "truth" and "lies" differ on a linguistic level. In spite of this, the assumption that a speaker will "focus on the objective (…) instead of using extraneous, distancing, or evasive measures" (Harpster, Adams, and Jarvis 2009: 78) is, unfortunately, very common.

Evaluation of the Claims

If we look at the claims collectively and in the context of the promotion of deception detection or veracity assessment methods, we will argue that the proponents of the claims are being epistemically *negligent*, because they have "a duty of epistemic care" but fail to "investigate in accordance with that duty" and fail to hedge their commitment to the claims (Heffer 2020: 202). In his definition of epistemic irresponsible discourses Heffer (2020) draws on philosopher Harry Frankfurt's notion of *bullshitting* which is a type of insincere speech which "deceives the hearer about the speaker's knowledge of what is said (the hearer falsely assumes that the speaker knows what she is talking about)" (Heffer 2020: 5). The present authors know that at least one of the proponents of Claims One–Five has been made aware that there is no scientific backing for their claims, meaning that they warrant the truth of what they say but conceal "the fact that [they have] no grounds for that warrant" (Heffer 2020: 5).

A Note on Pennebaker (2013)

One of the main issues that arises when dealing with these types of pseudolinguistic claims is that it can be difficult to determine their origins. In connection with this study, one of the authors familiarized herself with Pennebaker's *The Secret Life of Pronouns* (2013) and found several claims about linguistic lie detection. We feel compelled to address this issue briefly here, because it appears from our survey of the deception detection marketplace that Pennebaker (2013) is a likely academic source used by many proponents of deception detection, credibility, and validity assessment tools. This is valuable information for forensic linguists who seek to examine their claims further.

In the sixth chapter of the book, titled "Lying Words", Pennebaker (2013) makes several assertions about linguistic deception detection that we can recognize from tools like SCAN and the Reid Technique (such as *use of pronouns*, *passive constructions*, *avoiding answering a question*, *cognitive words*, etc.). For example, in a sub-section titled "The language of lying, liars, and scoundrels: in search of the linguistic lie detector" Pennebaker (2013: 140) states that it is theoretically possible to develop a linguistic lie detector tool and later adds: "not surprisingly, the language of lies is generally seen in the function words we use" (Pennebaker 2013: 169). This is a concerning statement given that the psychological experiments he uses to support that claim have low accuracy rates (51–73%), as referenced by himself.

Despite the large body of scientific evidence that refutes the assertions made about linguistic deception detection in *The Secret Life of Pronouns* (2013), tools like SCAN and the Reid Technique continue to be used. Although Pennebaker (2013) does not make these claims in the context of promoting specific lie detection tools, he has (perhaps unwittingly) laid a non-linguistic foundation for those interested in developing and using such tools.

Conclusion: Ethical Considerations When Addressing Pseudolinguistics

When confronted with pseudoscience, we face the simultaneous challenge of defending the validity of our own scientifically grounded approach, while addressing and countering misleading claims. To counter misinformation and maintain public trust, it is crucial to engage in effective science communication (Boudry et al. 2015). As argued above, forensic linguists have an ethical responsibility to identify and publicly illuminate the rift between evidence-based forensic linguistics and pseudolinguistic practices within law enforcement. As many other scientific fields are

discovering, the task of educating decision-makers (to say nothing of the general public) is often easier said than done.[1] As the medical scientists Capuano and Killu (2020: 246) note, "there are many cases in which providing education about pseudoscientific practice is insufficient to dissuade a caregiver from pursuing these treatments". This same kind of resistance can be found in the area of deception detection. As Heydon (2019: 63) explains, forensic linguistic research has produced a

> substantial and authoritative corpus of research literature dating back several decades that challenges the validity of lie detection, linguistic or otherwise, [which] seems to have had a depressingly small impact on the use of such methods in law enforcement and the justice system more broadly.

Although effective science communication is always a challenge, it is particularly so when researchers are faced with countering popular pseudoscientific theories, methods, and tools. The best weapon for fighting these faulty beliefs and practices remains education. Part of the ethical responsibility of the forensic linguist is to explain the true complexities of language use, the necessities of applying rigorous, transparent scientific methods, and the importance of insisting upon evidence-based findings as the basis for real-world policies. Another part is to address and preferably also correct faulty analyses promoted to sell tools and training that rely on language evidence. We urge that such corrections are offered in a respectful and constructive manner; no one is convinced by a competing account if they are shamed into doing it. In essence, when addressing pseudolinguistic claims in the public sphere, the forensic linguist will have to balance advocacy and objectivity. Admittedly, accomplishing these goals is no easy task, particularly when one considers that pseudoscientific claims about language-based deception detection are often also held by individuals and organizations in the criminal justice system. However, this fact does not alter the ethical imperative that forensic linguists be at the forefront of countering pseudoscientific claims about language and language users, since a failure to do so can have devastating consequences in terms of wrongfully convicting the innocent and allowing the guilty to go free.

Note

1 Many branches of science are familiar with pseudoscientists and their tendency "to reject results that are widely accepted by specialists working within the pertinent area of science" (Cleland and Brindell 2013: 183). There are a number of public controversies over scientific claims "in which scientists attempt to justify hypotheses and theories" (Cleland and Brindell 2013: 183), but fail to settle the discussion publicly after the initial damage has been done. A well-known example from the world of medicine is the debate about whether vaccines can cause autism in children. Although this claim has long since been debunked, it is still a matter of public debate (Taylor, Swerdfeger, and Eslick 2014).

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